$$\tau_z\sigma_z\frac{1}{\sqrt{2}}\Big(|ud\rangle - |du\rangle\Big) = \frac{1}{\sqrt{2}}\Big(-|ud\rangle + |du\rangle\Big).$$

Notice that the end result is just to change the sign of |*sing*>:

$$\tau_z\sigma_z|sing\rangle = -|sing\rangle.$$

Evidently, |*sing*> is an eigenvector of the observable $\tau_z\sigma_z$ with eigenvalue −1. Let's examine the significance of this result. Alice measures σ_z and Bob measures τ_z; when they come together and compare results, they find they've measured opposite values. Sometimes, Bob measures +1 and Alice measures −1. Other times, Alice measures +1 and Bob measures −1. The product of the two measurements is always −1.

There should be nothing surprising in this result. The state-vector |*sing*> is a superposition of two vectors, |*ud*> and |*du*>, both of which comprise two spins with opposite *z* components. The situation is altogether similar to the classical example involving Charlie and his two coins.

But now we come to something that has no classical analog. Suppose that instead of measuring the *z* components of their spins, Alice and Bob measure the *x* components. To find out how their outcomes are correlated, we must study the observable $\tau_x\sigma_x$.

Let's act on |*sing*> with this product. Here are the steps:

$$\begin{aligned}\tau_x\sigma_x|sing\rangle &= \tau_x\sigma_x\frac{1}{\sqrt{2}}\Big(|ud\rangle - |du\rangle\Big)\\ &= \tau_x\frac{1}{\sqrt{2}}\Big(|dd\rangle - |uu\rangle\Big)\\ &= \frac{1}{\sqrt{2}}\Big(|du\rangle - |ud\rangle\Big)\end{aligned}$$